



Faculty of Computer Science and Information Technology

***DIGITAL MANAGEMENT SYSTEM FOR PERSATUAN KENYAH
BADENG SARAWAK (KEBANA)***

Cedric Hilary Samah (86350)

Bachelor of Computer Science with Honours
(Information Systems)
2025

**DIGITAL MANAGEMENT SYSTEM FOR PERSATUAN KENYAH BADENG
SARAWAK (KEBANA)**

CEDRIC HILARY SAMAH

This project is submitted in partial fulfilment of the
requirements for the degree of
Bachelor of Computer Science and Information Technology

Faculty of Computer Science and Information Technology
UNIVERSITI MALAYSIA SARAWAK
2025

ACKNOWLEDGEMENT

First and foremost, I would like to express my deepest gratitude to my supervisor, Dr. Chew Kim Mey, for their invaluable guidance, patience, and continuous support throughout the completion of this Final Year Project. Their technical advice and constructive feedback were instrumental in shaping the direction of this project and ensuring its alignment with academic standards. I would also like to extend my sincere appreciation to the committee members of Persatuan Kenyah Badeng Sarawak (KEBANA), particularly the Secretary and Treasurer. Their cooperation during the data gathering phase and their willingness to share detailed insights into their operational challenges were crucial in designing a system that truly meets the community's needs.

Special thanks go to the Faculty of Computer Science and Information Technology, Universiti Malaysia Sarawak (UNIMAS), for providing the necessary resources and learning environment to undertake this study. Finally, I would like to thank my family and friends for their encouragement, understanding, and emotional support during the development of this project. Their belief in my abilities provided me with the motivation to overcome the challenges encountered along the way.

Table of Contents

ACKNOWLEDGEMENT	ii
TABLE OF FIGURES.....	vii
TABLE OF TABLES	viii
LIST OF ABBREVIATIONS	ix
ABSTRACT	x
<i>ABSTRAK</i>	xi
CHAPTER 1: INTRODUCTION	1
1.1 Introduction	1
1.2 Problem Statement	1
1.3 Aims & Objectives	2
1.4 Scope	3
1.5 Methodology	4
1.6 Significance of Project	4
1.7 Project Schedule	5
1.8 Expected Outcome	5
CHAPTER 2: LITERATURE REVIEW	6
2.1 Introduction	6
2.2 Overview of Digital Transformation in Community Organizations	8

2.2.1	The Role of Digital Strategy in NGO Efficiency	8
2.2.2	Digital Transformation in the Malaysian Rural Context	8
2.2.3	Alignment with Sarawak’s Post COVID-19 Development Strategy (PCDS) 2030.....	9
2.3	Review of Similar Existing Systems	9
2.3.1	Commercial Solution: WildApricot	10
2.3.2	Open-Source Solution: CiviCRM	10
2.3.3	Current Method: Microsoft Excel & Manual Records	11
2.4	Comparison of Existing Systems	12
2.5	Analysis of Gaps (The Problem with Existing Solutions)	13
2.6	Proposed Features for KEBANA Digital Management System.....	14
2.6.1	Digital Membership Registry (Internal Database)	14
2.6.2	Event Proposal and Document Archiving	15
2.6.3	Internal Financial Ledger	15
2.6.4	Role-Based Access Control (RBAC)	16
CHAPTER 3: METHODOLOGY		17
3.1	Introduction	17
3.2	Software Development Methodology	17
3.2.1	Justification for Choosing Agile	18
3.2.2	The Development Phases	19

3.3	Requirement Analysis	20
3.3.1	Functional Requirements	20
3.3.2	Non-Functional Requirements	22
3.4	System Design.....	23
3.4.1	Use Case Design.....	23
3.4.2	Activity Diagram (Process Flow)	25
3.4.3	Entity Relationship Diagram (ERD)	26
3.4.4	User Interface Design	28
3.5	System Development Environment.....	29
3.5.1	Hardware Requirements	29
3.5.2	Software Requirements.....	29
3.6	System Testing Plan	30
3.6.1	Unit Testing.....	30
3.6.2	User Acceptance Testing (UAT)	31
3.7	Chapter Summary.....	31
CHAPTER 4: EXPECTED RESULTS & DISCUSSION		32
4.1	Introduction	32
4.2	Expected Outcomes.....	32
4.3	Discussion of Project Significance.....	33

4.4	Potential Implementation Challenges.....	34
4.5	Conclusion.....	34
REFERENCES		36
APPENDICES.....		38
1.	Letter of Intent.....	38

TABLE OF FIGURES

Figure 1.1	Gantt Chart of FYP 1 Project Schedule.....	5
Figure 3.1	Iterative Feedback Loop Flow.....	18
Figure 3.2	Use Case Diagram of the KEBANA Digital Management System.....	23
Figure 3.3	Event Proposal Creation.....	25
Figure 3.4	ERD Diagram.....	26

TABLE OF TABLES

Table 2.1	Comparison of Existing Systems vs. Proposed Systems.....	12
Table 3.1	Functional Requirements of the Digital Management System for the KEBANA...	20
Table 3.2	Non-Functional Requirements of the Digital Management System for the KEBANA.....	22

LIST OF ABBREVIATIONS

Abbreviation	Full Term
AGM	Annual General Meeting
CRUD	"Create, Read, Update, and Delete"
CSS	Cascading Style Sheets
ERD	Entity Relationship Diagram
FK	Foreign Key
FYP	Final Year Project
HTML	HyperText Markup Language
IC	Identity Card
IDE	Integrated Development Environment
IT	Information Technology
KEBANA	Persatuan Kenyah Badeng Sarawak
NGO	Non-Governmental Organization
ORM	Object-Relational Mapping
PaaS	Platform as a Service
PBKDF2	Password-Based Key Derivation Function 2
PCDS	Post COVID-19 Development Strategy
PDF	Portable Document Format
PK	Primary Key
RBAC	Role-Based Access Control
RDBMS	Relational Database Management System
SDLC	Software Development Life Cycle
SSOT	Single Source of Truth
UAT	User Acceptance Testing
UI	User Interface
VS Code	Visual Studio Code
XSS	Cross-Site Scripting

ABSTRACT

The rapid advancement of digital technology has created a significant divide between urban and rural organizations in managing administrative operations. The Persatuan Kenyah Badeng Sarawak (KEBANA), a rural community association, currently relies on manual, paper-based methods and decentralized spreadsheets to manage its membership, events, and finances. This traditional approach has led to critical issues such as data redundancy, loss of institutional memory, and a lack of financial transparency. In alignment with the Sarawak Post COVID-19 Development Strategy (PCDS) 2030, which emphasizes social inclusivity and digital transformation, this project aims to develop a Web-Based KEBANA Management System.

The proposed system is designed as a centralized administrative tool strictly for the use of the KEBANA committee. It features three core modules: a Membership Database to serve as a single source of truth for member records; an Event Management module for archiving proposals and meeting minutes; and a Financial Tracking module to automate the generation of monthly statements. The system is developed using the Agile Methodology, ensuring iterative feedback from the committee, and utilizes Python (Django) and MySQL to ensure security and scalability. A key design priority is "Low-Cost Sustainability," ensuring the system can be deployed on cost-effective cloud infrastructure. The expected outcome is a fully functional, mobile-responsive system that significantly reduces administrative workload, enhances data security, and empowers the KEBANA committee to operate with greater efficiency and transparency.

ABSTRAK

Kemajuan pesat teknologi digital telah mewujudkan jurang yang ketara antara organisasi bandar dan luar bandar dalam menguruskan operasi pentadbiran. Persatuan Kenyah Badeng Sarawak (KEBANA), sebuah persatuan komuniti luar bandar, pada masa ini bergantung kepada kaedah manual berasaskan kertas dan hamparan elektronik yang tidak berpusat untuk menguruskan keahlian, acara, dan kewangan mereka. Pendekatan tradisional ini telah mengakibatkan isu-isu kritikal seperti pertindihan data, kehilangan rekod institusi, dan kurangnya ketelusan kewangan. Selaras dengan Strategi Pembangunan Pasca COVID-19 Sarawak (PCDS) 2030, yang menekankan keterangkuman sosial dan transformasi digital, projek ini bertujuan untuk membangunkan Sistem Pengurusan KEBANA berasaskan laman sesawang.

Sistem yang dicadangkan ini direka bentuk sebagai alat pentadbiran berpusat khusus untuk kegunaan jawatankuasa induk KEBANA. Ia menampilkan tiga modul teras iaitu, Pangkalan Data Keahlian yang berfungsi sebagai sumber rujukan utama (single source of truth) bagi rekod ahli, modul Pengurusan Acara untuk mengarkibkan kertas kerja cadangan dan minit mesyuarat, serta modul Penjejakan Kewangan untuk mengautomasikan penjanaan penyata bulanan. Sistem ini dibangunkan menggunakan metodologi Agile (Agile Methodology), bagi memastikan maklum balas iteratif daripada jawatankuasa, serta menggunakan Python (Django) dan MySQL untuk menjamin keselamatan dan kebolehskalaan. Keutamaan utama reka bentuk adalah "Kelestarian Kos Rendah", bagi memastikan sistem ini boleh ditempatkan pada infrastruktur awan yang kos efektif. Hasil yang dijangkakan adalah sebuah sistem yang berfungsi sepenuhnya dan responsif, yang dapat mengurangkan beban kerja pentadbiran secara signifikan, meningkatkan keselamatan data, dan memperkasakan jawatankuasa KEBANA untuk beroperasi dengan lebih cekap dan telus.

CHAPTER 1: INTRODUCTION

1.1 Introduction

Effective information management is crucial for a large community. Delivering concise and accurate information is essential to avoid miscommunication, preserve harmony, and ensure no one is left behind in this digital journey. In Sarawak, organizations like the Persatuan Kenyah Badeng Sarawak (KEBANA) have a large and active membership, which presents a significant management challenge. Currently, the committee relies on manual, paper-based, or decentralized methods for record-keeping and communication. This can lead to administrative inefficiencies and difficulty in delivering information effectively.

To address these challenges, this project proposes the design and development of a KEBANA Digital Management System. This web-based platform will provide a centralized, secure, and efficient tool for all KEBANA members, and especially for the KEBANA committee. It will streamline operations, improve data accuracy, and provide a single, official channel for all communication.

1.2 Problem Statement

The Kenyah Badeng National Association (KEBANA) serves a widespread community originating from numerous remote villages in Sarawak, primarily in the Belaga and Ulu Baram areas. Annually, the association organizes the "Ramai Biu' KEBANA" festival, a significant cultural event held in both rural villages and major towns. However, the planning and management of this festival face significant operational challenges. The organizing committee currently relies on traditional, manual processes for essential administrative functions, including event

documentation, the creation of proposals, and the generation of financial reports. This reliance on manual methods is fundamentally inefficient and unsustainable for two key reasons:

1. Geographical Dispersion where made the committee members are not centrally located but are scattered throughout Malaysia.
2. Lack of Centralization is shown where there is no single, accessible platform for collaboration and data management.

Consequently, this manual approach leads to considerable administrative overhead, a high risk of data loss or inconsistency, and significant delays in decision-making, ultimately compromising the efficiency and success of their most important community event.

1.3 Aims & Objectives

This project aims to achieve the following objectives:

1. To design and develop an event management module that allows KEBANA committee members to create events, manage proposals, and archive related documentation in a centralized online repository.
2. To implement a financial management module capable of tracking income and expenditures and generating automated financial reports to ensure accuracy and transparency for auditing purposes.
3. To create a centralized and searchable database for managing the personal and membership details of all KEBANA members, replacing the current manual record-keeping methods.

1.4 Scope

The KEBANA Management System will be developed as a centralized web-based application designed to streamline the committee's daily operations. The system's primary function is to serve as a comprehensive database for membership management. It will allow the committee to securely store and search for member details such as names, identification numbers, and contact information. Authorized users will be able to add new members, update existing records, and deactivate old accounts, while general members will have view-only access to verify their own personal details.

In addition to membership data, the system will feature a dedicated module for event and proposal management. This will enable the committee to digitize the event planning process by creating new events with details like date, time, and venue directly in the system. It will also serve as a digital repository, allowing the committee to write proposals and archive important documents, such as approval letters and photos, ensuring that all event-related information is stored safely in one place for future reference.

To support financial transparency, the system will include a financial management module for tracking the association's income and expenditures. The Treasurer will be able to record transactions and categorize them into groups such as hall rentals, donations, or food and beverage costs. The system will then use this data to automatically calculate totals and generate basic financial statements, which will assist the committee in auditing and maintaining accurate financial records.

To ensure the project is completed within the academic timeframe, certain complex features will be excluded from the scope. The system will not include an online payment gateway for processing real-time bank transfers, nor will it handle complex accounting tasks like tax calculations. Furthermore, the project will not involve the development of a native mobile application for Android or iOS, and it does not cover the manual data entry of historical paper records into the new database.

1.5 Methodology

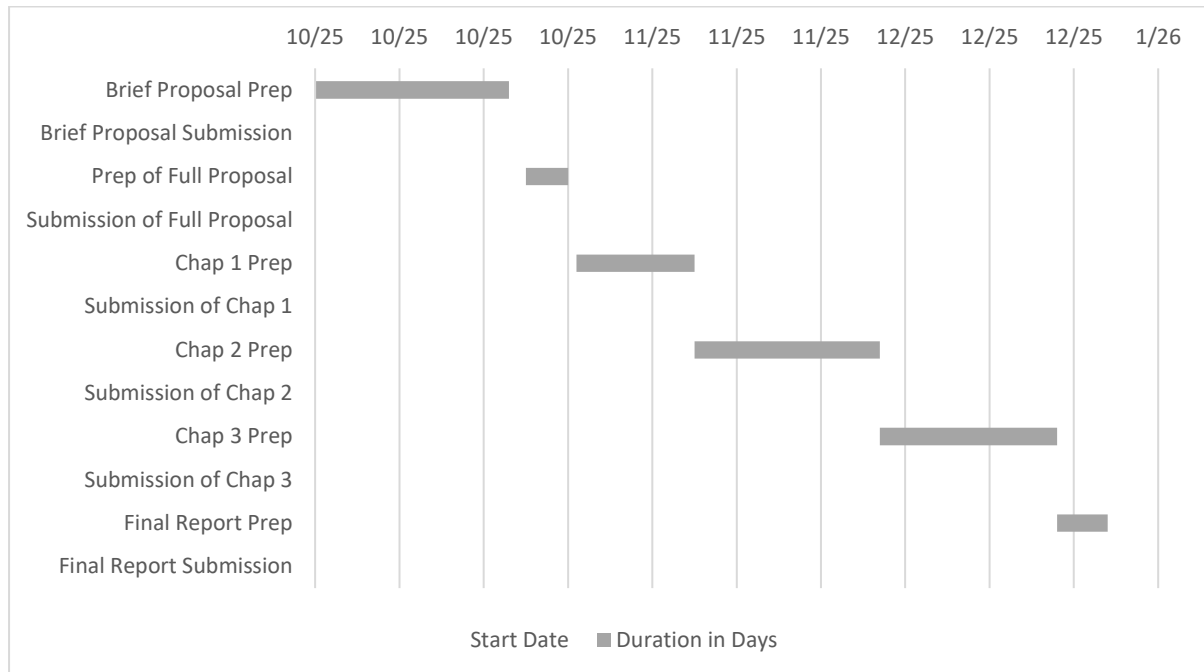
Agile Methodology has been selected for this project to facilitate close collaboration with the KEBANA committee. This approach is essential for ensuring the system's development aligns with their organizational needs. Given that this will be the committee's first digital management system, the flexibility of Agile is a significant advantage. It allows for an iterative development process where requirements can evolve, and stakeholder feedback can be incorporated continuously.

1.6 Significance of Project

The KEBANA Digital Management System is a significant digital transformation project, replacing the association's manual, paper-based methods with a single, centralized, and efficient digital platform. This system will streamline the committee's administrative operations and establish a "single source of truth" for all members and event data, improving governance and continuity. It will also encourage a more engaged community by providing members with a central hub for official announcements, event calendars, and personal profile management. Academically, this project serves as a practical application of data engineering and Agile principles to solve a tangible, real-world community problem.

1.7 Project Schedule

Figure 1.1 Gantt Chart of FYP 1 Project Schedule



1.8 Expected Outcome

Upon the successful completion of this project, the primary outcome will be a functional, deployed, web-based KEBANA Digital Management System. This system will be fully operational and ready for use by the KEBANA committee and its members, providing a single, centralized platform that successfully digitizes their core administrative functions. This includes the modules for member management, event and budget planning, announcements, and document sharing as defined in the project scope.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The landscape of community development in Sarawak is undergoing a significant paradigm shift, driven by the state's aggressive pursuit of modernization and digital adoption. This chapter reviews the literature related to digital transformation in non-governmental organizations (NGOs) and community associations, grounding the discussion within the overarching framework of the Sarawak Post COVID-19 Development Strategy (PCDS) 2030.

The PCDS 2030 serves as the primary roadmap for the state, articulating a vision where, "By 2030, Sarawak will be a thriving society driven by data and innovation where everyone enjoys economic prosperity, social inclusivity and sustainable environment". This vision is anchored on three pillars: Economic Prosperity, Inclusive Society, and Environmental Sustainability. Crucially, the strategy identifies Digital Transformation not merely as a goal, but as a critical "enabler" that supports all economic sectors. The state aims to empower its economic sectors to increase efficiency and productivity through this transformation, ensuring that the benefits of development are distributed equitably.

For community organizations like the Persatuan Kenyah Badeng Sarawak (KEBANA), the PCDS 2030's emphasis on Social Inclusivity is particularly relevant. The strategy explicitly aims to bridge the developmental gap between urban and rural areas. It outlines the need for "greater urban rural economic integration" and "community participation in mainstream development".

This involves capacity building for communities through skills training and entrepreneurship, ensuring they have the digital connectivity and tools to participate in the digital economy. Without adopting digital tools, rural communities risk deepening the "digital divide" that separates them from urban populations.

The literature further suggests that for NGOs to remain relevant and effective in this new era, they must adopt specific digital strategies. A cohesive digital plan allows an NGO to "fulfill its mission faster and easier, to save and use resources as rationally as possible," and to increase interaction with stakeholders. Conversely, a lack of digital strategy can lead to "loss of opportunities for community development" and hinder cooperation. Studies on digital transformation in Malaysia and Indonesia highlight that integrating technology into rural development has already demonstrated enhanced economic productivity within those communities. Therefore, the alignment of KEBANA's operations with digital standards is not just an administrative upgrade; it is a strategic necessity to ensure the community is not left behind in Sarawak's journey toward 2030.

This chapter will further explore existing systems and the specific benefits of digitalization for community management, establishing the gap that the proposed KEBANA Management System intends to fill.

2.2 Overview of Digital Transformation in Community Organizations

The integration of digital technologies into the non-profit and community sector is no longer an optional luxury but a critical necessity for sustainability and growth. Digital transformation in this context is defined as an irreversible process where organizations must adapt to technological trends to increase performance and resilience. For non-governmental organizations (NGOs) and community associations, a coherent digital strategy is a vital tool that allows them to fulfill their mission faster, utilize resources more rationally, and significantly increase interaction with stakeholders.

2.2.1 The Role of Digital Strategy in NGO Efficiency

The primary principle for digital adoption in NGOs is operational efficiency. Bălăcescu (2021) highlights that reliance on manual, paper-based methods is costly and prone to human error. For instance, managing volunteer or member files on paper makes it "almost impossible to identify the necessary information... at the right time". In contrast, digital solutions provide real-time information and ensure total transparency of actions, allowing stakeholders to be directly involved in decision-making processes. Furthermore, the lack of a digital strategy can lead to lost opportunities for community development and cooperation.

2.2.2 Digital Transformation in the Malaysian Rural Context

In the specific context of Malaysia, digital transformation has been a key focus for rural development. Haron et al. (2023) note that the rapid proliferation of Information and Communication Technology (ICT) has been instrumental in bridging the digital divide between rural and urban communities. Initiatives such as the establishment of telecentres in rural areas have

introduced the digital realm to these communities, providing access to online facilities and global perspectives. The application of ICT in administrative processes has led to "heightened efficiency," where transactions are streamlined, resulting in time savings and cost reductions for both individuals and organizations. However, challenges remain; specifically, the transition to new technology requires adaptation, which can be a significant hurdle for elderly residents or those with limited digital literacy.

2.2.3 Alignment with Sarawak's Post COVID-19 Development Strategy (PCDS) 2030

The push for digitalization in community organizations is strongly supported by state-level policy. The Post COVID-19 Development Strategy 2030 (PCDS 2030) outlines Sarawak's aspiration to become a developed state driven by "data and innovation". The strategy identifies "Digital Transformation" as a key enabler to empower economic sectors and increase efficiency. Crucially, the PCDS 2030 emphasizes "Social Inclusivity" as a core pillar, with a specific objective to narrow the developmental gap between urban and rural areas. The strategy advocates "greater urban rural economic integration" supported by digital connectivity, ensuring that rural communities have equal opportunities to participate in the mainstream economy. For community associations like KEBANA, adopting a digital management system is a direct contribution to this state vision, moving the community towards a "digital society" that is resilient and inclusive.

2.3 Review of Similar Existing Systems

To understand the functional requirements of a digital management system, this study reviewed three categories of existing solutions: commercial software, open-source platforms, and the traditional manual methods currently employed by many rural organizations.

2.3.1 Commercial Solution: WildApricot

WildApricot is widely regarded as the leading membership management software for small to medium-sized associations. It offers an “all-in-one” platform that includes a website builder, event registration, automated membership renewals, and financial tracking (GroupApp, 2025).

The advantage of WildApricot is it is highly comprehensive, allowing for complex event ticketing and automated email reminders for dues. It also offers a mobile app for admins to manage the organization on the go.

The disadvantage of it is when it comes to applying it to the KEBANA community. This is when the cost becomes the barrier. Pricing starts at approximately \$60 USD (~RM 280) per month for just 100 contacts, which is prohibitively expensive for a community-funded association. Furthermore, the interface is complex, designed for users with a higher level of digital literacy, and features like payment processing are tailored for US/Western bankin systems (Stripe/PayPal) rather than local Malaysian transfers.

2.3.2 Open-Source Solution: CiviCRM

CiviCRM is a powerful, open-source Constituent Relationship Management (CRM) system designed specifically for non-profits and NGOs. It provides extensive features for managing members, donors, and case management (CiviCRM, 2025).

In terms of the advantage, as an open-source software, the license is free. It is highly customizable and can manage complex data relationships, such as households or village groups.

Even so, the disadvantage of this software is that CiviCRM has a steep technical learning curve. It requires a dedicated server and significant technical expertise to install, configure, and maintain. For an organization like KEBANA, which lacks a dedicated IT department, the “technical overhead” makes this solution unsustainable in the long run.

2.3.3 Current Method: Microsoft Excel & Manual Records

Currently, many rural associations rely on general-purpose tools like Microsoft Excel and WhatsApp for management. While these tools are familiar, free, and require no setup, they also have significant weaknesses where this method suffers from severe data integrity and security issues. As noted in the literature, “managing data across multiple spreadsheets leads to duplication and errors” (MemberJungle, 2023). There is no “single source of truth,” meaning if one committee members updates a file, others may still be using an old version. Furthermore, sharing sensitive member data (like IC numbers) via WhatsApp poses a significant security risk.

2.4 Comparison of Existing Systems

The following table summarizes the comparison between the market leaders and the proposed KEBANA Digital Management System (KDMS).

Table 2.1 Comparison of Existing Systems vs. Proposed Systems

Feature / Criteria	WildApricot (Commercial)	CiviCRM (Open Source)	Microsoft Excel (Current Method)	Proposed KDMS
Cost	High (Recurring Monthly Fee)	Free License	Low / Free	<i>(To-be discussed)</i>
User Friendliness	Medium (Complex menus)	Low (Requires technical training)	High (Familiar)	High (Tailored Minimalist UI)
Data Centralization	Yes	Yes	No (Files scattered)	Yes (Single Database)
Security & Access	High (Role-based)	High (Role-based)	Low (File sharing risks)	High (Secure Login)
Financial Module	Advanced (Complex Accounting)	Advanced	Manual Calculation	Simplified (Income/Expense Tracking)

Event Focus	Ticketing & Registration	Fundraising & Campaigns	Basic List	Proposal & Planning Archiving
Language & Context	English (US-Centric)	English (Global)	Any	English, Bahasa Malaysia, Kenyah (Highly customizable localized for Sarawak)

2.5 Analysis of Gaps (The Problem with Existing Solutions)

The review highlights a significant gap in the current market. Commercial tools like WildApricot are too expensive and complex for a community association operating on a limited budget. Open-source tools like CiviCRM, while free, require technical maintenance that exceeds KEBANA's capacity. Meanwhile, the current manual method fails to provide the security, efficiency, and data centralization required by the *Sarawak Post COVID-19 Development Strategy 2030*.

Therefore, there is a clear need for a custom-developed, lightweight web application. This system must bridge the gap by offering the centralization and security of professional software, but with the simplicity and low cost required by a rural community organization. It is important to note that while 'Open Source' software like CiviCRM is technically free to download, the total cost of ownership is high due to the need for powerful servers and professional IT maintenance. In contrast, the proposed KEBANA Management System is designed to be lightweight.

It can be hosted on entry-level cloud platforms (costing approximately RM 100–RM 200 per year for domain and hosting), making it a financially sustainable option for the association's long-term budget.

2.6 Proposed Features for KEBANA Digital Management System

Based on the review of existing systems and the specific operational needs of the KEBANA committee, this project proposes the development of an KEBANA *internal* Digital Management System. Unlike commercial software that focuses on public-facing features (like ticketing or social networking), this system is designed strictly as a backend tool to digitize the committee's manual workflows. The system will incorporate the following key features to address the identified limitations.

2.6.1 Digital Membership Registry (Internal Database)

To replace the decentralized Excel files currently used by the Secretary, the system will provide a secure, *committee-only database*. This features a comprehensive CRUD (Create, Read, Update, Delete) interface that allows the Secretary, or any other allowed committee members, to maintain accurate records of all KEBANA members. Furthermore, the system will also focus on high-speed data entry and search functionality for the committee, rather than a self-service portal for members. This eliminates the need for managing thousands of member passwords and significantly reduces cybersecurity risks. In terms of Data Integrity, the system ensures a “Single Source of Truth” (SsoT), preventing the issue of duplicate or outdated member lists circulating among committee members.

2.6.2 Event Proposal and Document Archiving

Recognizing that KEBANA's event management is administrative rather than commercial, this module focuses on the planning and approval phase. It offers a *digital repository* where the Committee can draft, edit, and store Event Proposals. In addition to that, *digital archiving* enables the Committee to upload and tag official documents (e.g. government approval letters, meeting minutes, post-moterm reports) to specific events. This preserves the association's institutional memory, solving the problem of lost physical files.

2.6.3 Internal Financial Ledger

To assist the Treasurer, the system will include a simplified financial tracking module designed for non-accountants. Its key feature is by providing a “Digital Cashbook” for recording inflow (e.g., membership fees, grants) and outflow (e.g., event logistics). It will also feature an Automated Reporting, where the system will automatically generate standard financial reports (e.g., Monthly Income & Expenditure Statement) required for committee meetings and the Annual General Meeting (AGM), ensuring transparency without the complexity of full accounting software.

2.6.4 Role-Based Access Control (RBAC)

To ensure data security within the committee, the system will implement strict access levels. Different committee roles will have tailored views. For example, the Treasurer may have exclusive written access to financial records, while the Secretary manages membership data. The President may have a “Super Admin” view to oversee all modules. By restricting login access exclusively to a small group of authorized committee members, the system remains lightweight and highly secure against external unauthorized access.

CHAPTER 3: METHODOLOGY

3.1 Introduction

This chapter outlines the systematic approach and technical framework adopted for the design and development of the KEBANA Digital Management System. The primary objective of this project is to transform the association's manual, paper-based administrative processes into a centralized web-based application. To ensure the successful delivery of this system, this chapter details the chosen software development life cycle (SDLC), the requirement analysis process, the system design architecture, and the testing strategies that will be employed. The methodology serves as a roadmap to guide the development from the initial concept to the final deployment. It ensures that the resulting system is not only technically robust but also user-friendly and sustainable for the KEBANA committee's specific operational needs.

3.2 Software Development Methodology

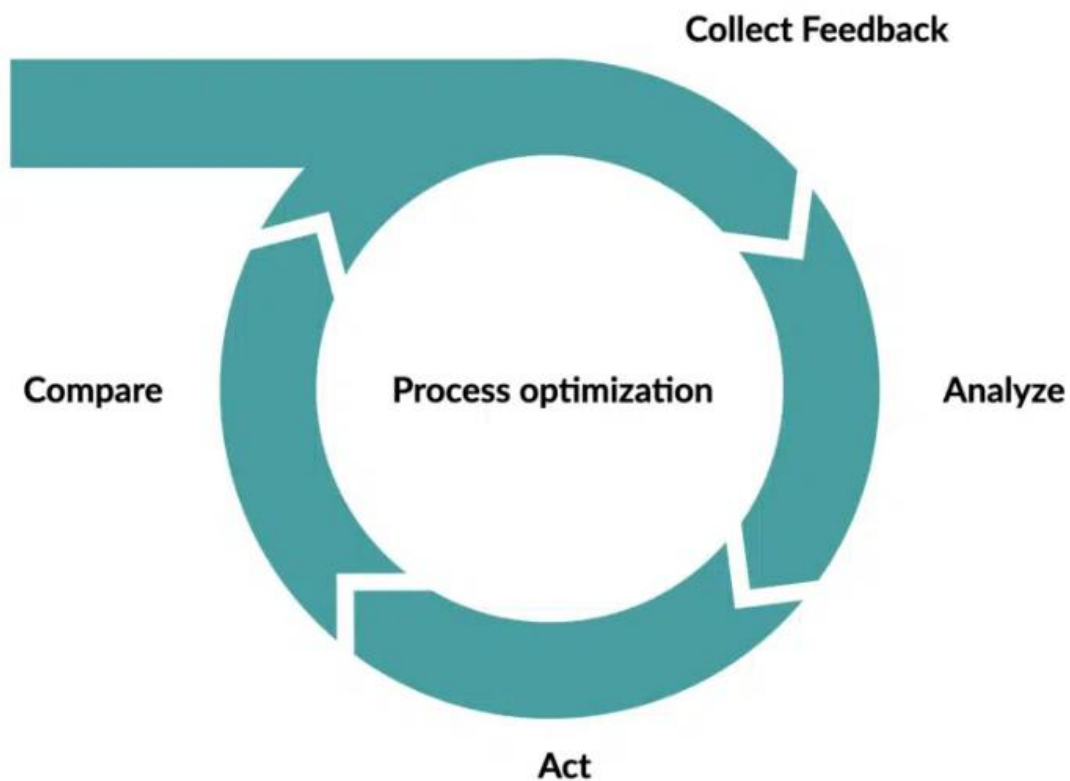
For the development of this system, the Agile Software Development Methodology has been selected. Unlike traditional models (such as Waterfall) which follow a rigid, linear path, Agile focuses on iterative development and continuous feedback.

3.2.1 Justification for Choosing Agile

The decision to adopt the Agile methodology is driven by three key factors specific to the KEBANA context.

1. Iterative Feedback Loop

Figure 3.1 Iterative Feedback Loop Flow



The KEBANA committee members have varying levels of digital literacy. By developing the system in small cycles (Sprints), the developer can demonstrate working modules, such as the Membership Database, to the committee early in the process. This allows for immediate feedback and adjustments before moving on to complex features like Financial Reporting.

2. Flexibility to Change

The requirements in community projects often evolve once users see the actual software. Agile allows flexibility in the project scope, enabling the developer to refine features based on real-world testing rather than strictly adhering to initial assumptions.

3. Risk Mitigation

By testing small components frequently, technical bugs or usability issues are identified and resolved early. This reduces the risk of delivering a final product that the committee finds difficult to use.

3.2.2 The Development Phases

The project will follow the standard Agile lifecycle, which is structured into five distinct phases to ensure systematic progression from concept to deployment. The first phase, Planning and Requirement Gathering, involves conducting in-depth interviews with key committee members, such as the Secretary and Treasurer. This phase is critical for understanding their specific manual workflows, identifying pain points, and defining the initial user stories. Following this, the System Design phase focuses on creating the architectural blueprints for the system. This involves developing Use Case Diagrams to map out user interactions and Entity Relationship Diagrams (ERD) to structure the database.

Once the design is finalized, the Iterative Development phase begins, where the system is coded in a series of “Sprints”. Each sprint focuses on building a specific functional block, such as the Membership Module, followed by the Event Management Module and the Finance Module.

After each module is developed, the project moves to the Testing and Review phase. This involves performing Unit Testing to ensure code quality and User Acceptance Testing (UAT) with the committee to gather feedback. Finally, the Deployment phase involves configuring the cloud hosting environment and formally handing over the fully functional system to the KEBANA committee for live operations.

3.3 Requirement Analysis

Following the initial consultation with the KEBANA committee, the system requirements have been categorized into Functional Requirements, specifically focuses on the behaviors and features, and Non-Functional Requirements, where they focus on the system attributes like security and performance.

3.3.1 Functional Requirements

The functional requirements define the specific interactions authorized users will have with the system. These are grouped by the three distinct user roles which are Super Admin, Secretary, and Treasurer, as shown in the following **Table 3.1**.

Table 3.1 Functional Requirements of the Digital Management System for the KEBANA

Category	Requirement ID	Requirement Statements
Authentication & Access Control Module	FR_01	The system shall allow users to log in using a secure username and password.
	FR_02	The system shall restrict access to specific modules based on the assigned user role (e.g., only the Treasurer can access the Finance module.
	FR_03	The system shall automatically log users out after a period of inactivity to prevent unauthorized access.

Membership Management Module	FR_04	The system shall allow the Secretary to Create new member profiles with fields for Name, IC Number, Village, and Contact Info.
	FR_05	The system shall allow the Secretary to Update existing member details (e.g., change of address or phone number).
	FR_06	The system shall allow the Secretary to Search for members using keywords (Name or IC Number) to quickly retrieve records.
	FR_07	The system shall validate the IC Number to prevent duplicate member registrations (Single Source of Truth)
Event & Document Management Module (Secretary Role)	FR_08	The system shall allow the creation of Event Proposals, capturing details such as Event Title, Date, Venue, and Estimated Budget.
	FR_09	The system shall provide a File Upload feature, allowing the Secretary to attach PDF or Image files (e.g., approval letters, meeting minutes) to a specific event record.
	FR_10	The system shall allow the retrieval and downloading of these archived documents for future reference.
Financial Management Module (Treasurer Role)	FR_11	The system shall allow the Treasurer to record Financial Transactions, categorizing them as either “Income” (e.g., Fees, Donations) or “Expenditure” (e.g., Food, Logistics).
	FR_12	The system shall automatically calculate the current Cash Balance based on the recorded transactions.
	FR_13	The system shall generate a printable Monthly Financial Statement (PDF) summarizing total income and total expenditure for the selected month.
System Administration (Super Admin Role)	FR_14	The system shall allow the Super Admin to create and delete user accounts for committee members.
	FR_15	The system shall allow the Super Admin to manage dropdown configurations (e.g., adding a new “Village” name to the selection list).

3.3.2 Non-Functional Requirements

These requirements define the quality attributes of the system, ensuring it is usable and sustainable for KEBANA. These requirements are listed in **Table 3.2**.

Table 3.2 Non-Functional Requirements of the Digital Management System for the KEBANA

Category	Requirement ID	Requirement Statements
Usability	NFR_01	The user interface (UI) shall be minimalist and intuitive, designed for users with basic digital literacy. It should use local terminology where appropriate to ensure clarity.
Mobile Responsiveness	NFR_02	The system should be fully responsive, adapting its layout for mobile phone screens (Android/iOS) to allow committee members to access data during meetings without a laptop.
Data Security	NFR_03	The system shall use hashing algorithms (e.g., PBKDF2 or Argon2) to store user passwords securely. Direct database access shall be restricted to the administrator only.
Performance	NFR_04	The system shall load the dashboard, and search results in under 5 seconds on a standard 4G mobile connection.
Sustainability	NFR_05	The system should be deployed on lightweight, cost-effective infrastructure (e.g., cloud platforms with free/low-cost tiers) to minimize the long-term financial burden on the association.

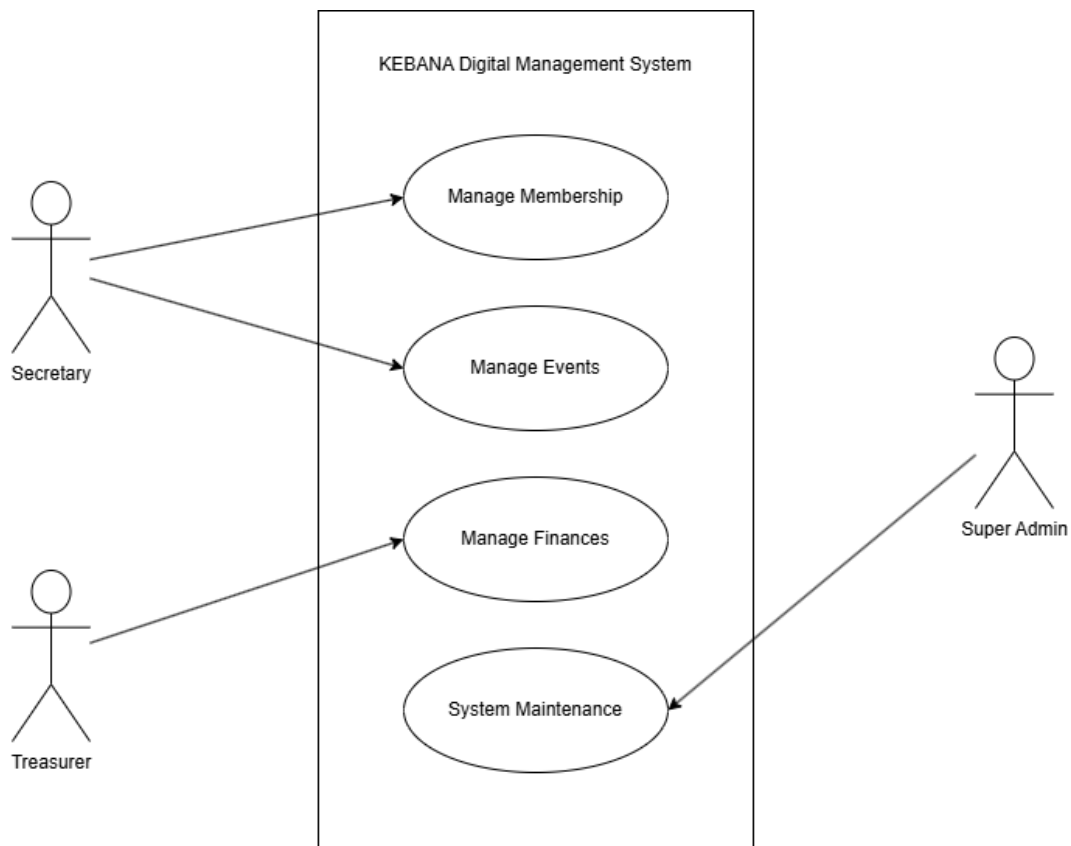
3.4 System Design

The system design phase translates the functional requirements into technical architecture. This section details the logical flow, data structure, and user interactions of the KEBANA Management System.

3.4.1 Use Case Design

The Use Case diagram visualises the behavioral aspect of the system, defining how the specific actors interact with the functional modules. As an internal administrative system, the actors are restricted to authorized committee members.

Figure 3.2 Use Case Diagram of the KEBANA Digital Management System

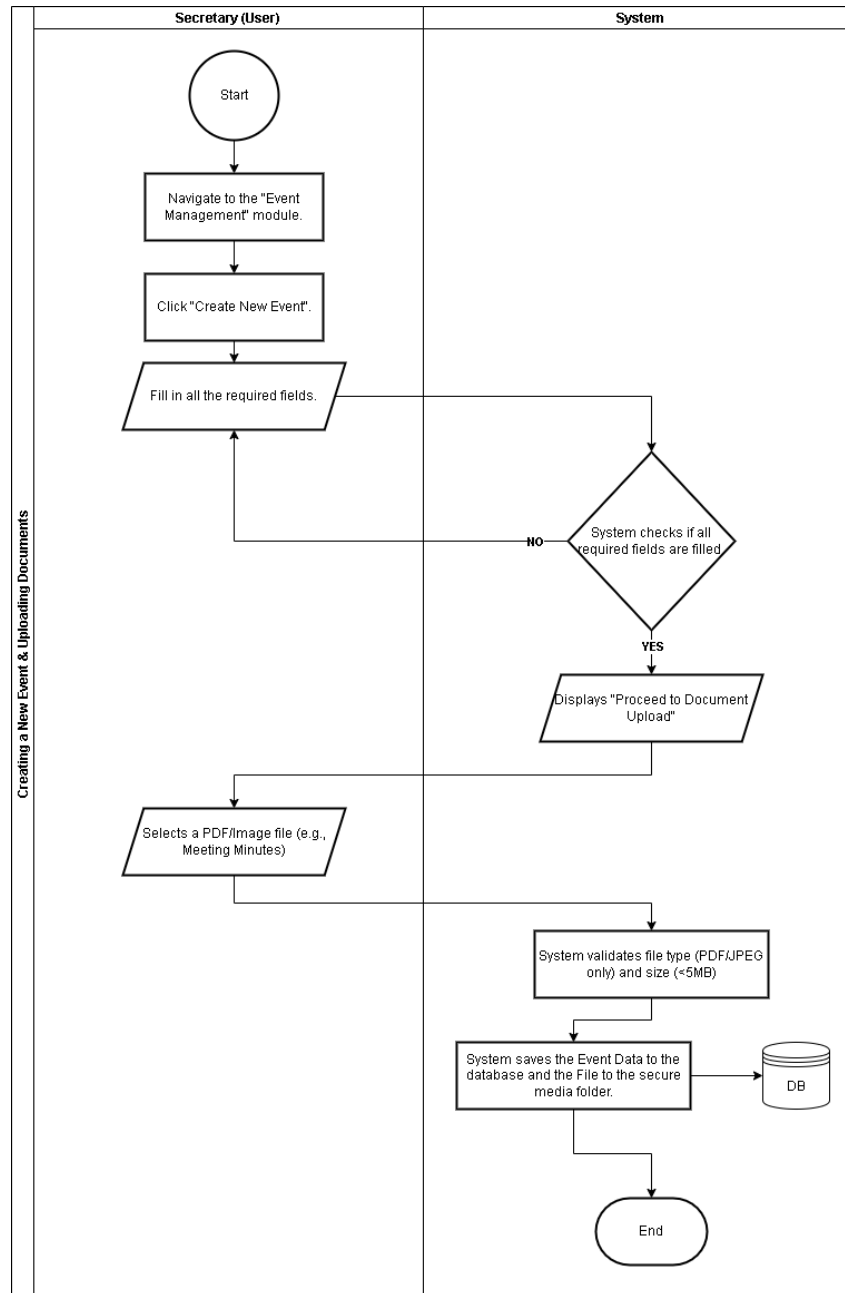


Based on Figure 3.2, it is shown that the Secretary role could Manage Membership. This role can Add, Update, Delete, and Search for KEBANA members. In addition to that, the Secretary could also Manage Events, by creating event proposals, and upload related digital documents such as minutes or letters. For the Treasurer role, it can record inflows/outflows and generate monthly financial statements. Lastly, for Super Admin role, it could create accounts for new committee members and manage dropdown configurations, such as List of Villages.

3.4.2 Activity Diagram (Process Flow)

To illustrate the system's logic, the Event Proposal Creation process is detailed below in Figure 3.3. This complex process involves data entry, file handling, and validation.

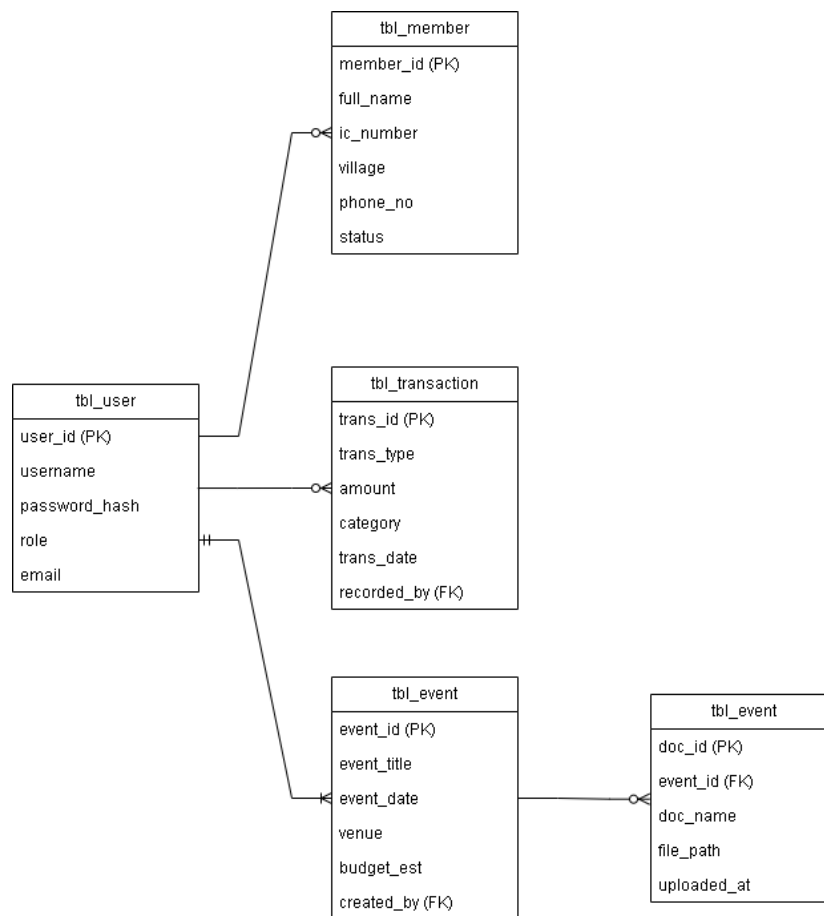
Figure 3.3 Event Proposal Creation



3.4.3 Entity Relationship Diagram (ERD)

The database architecture is designed using a Relational Model to ensure data integrity and minimize redundancy. The system consists of four primary entities that store the committee's operational data. **Figure 3.4** defines how these tables interact. The **User table** has a "One-to-Many" relationship with the Member, Event, and Transaction tables, meaning one committee member can create many records. Crucially, the **Event table** has a "One-to-Many" relationship with the **Document table**. This means that a single event (the "One" side) can have multiple uploaded documents (the "Many" side) attached to it, creating a structured digital filing system.

Figure 3.4 ERD Diagram



3.4.3.1 The User Entity (tbl_user)

This table stores the login credentials for the committee members. It acts as the central entity because every action in the system, whether adding a member or recording a transaction, must be performed by a logged-in user. It contains the user's secure password hash and their specific role (Admin, Secretary, or Treasurer).

3.4.3.2 The Member Entity (tbl_member)

This table serves as the "Single Source of Truth" for the association's membership records. It stores the personal details of the community members, such as their full name, IC number, and village. This table is linked to the User table to track which Secretary was responsible for registering the member.

3.4.3.3 The Event Entity (tbl_event)

This table records the details of the association's activities, such as the Annual General Meeting or Family Day. It captures the basic logistics of the event, including the event title, date, venue, and estimated budget.

3.4.3.4 The Document Entity (tbl_document)

To fulfill the requirement for "Digital Archiving," this table allows the system to store multiple files for a single event. For example, one event record might need to store a "Proposal Letter," an "Approval Letter," and a "Post-Mortem Report." By separating documents into their own table, the system allows the Secretary to upload an unlimited number of files for any given event.

3.4.3.5 The Transaction Entity (tbl_transaction)

This table is critical for the Financial Module. It records every inflow and outflow of funds managed by the Treasurer. Each record captures the transaction type (Income or Expense), the amount, the date, and the specific category (e.g., "Donation" or "Catering Cost"). This ensures that the system can automatically calculate the current cash balance and generate accurate monthly reports.

3.4.4 User Interface Design

The system design prioritizes mobile devices to match the daily needs of the committee. We use the Bootstrap 5 framework to ensure the website adjusts smoothly to different screen sizes. This allows the committee members to access the system using smartphones or tablets during their meetings in the village hall. They do not need to rely on a desktop computer or a separate mobile app to view the records.

The navigation menu features a sidebar that changes based on the device. On large screens, the menu remains visible for quick access to all modules. On small mobile screens, the menu hides inside a small icon to save valuable screen space. This simple layout helps users find what they need without feeling overwhelmed by too many buttons or complex options.

The main dashboard acts as a central command center for the users. It displays important information like the total number of members and the current financial balance in large and clear boxes. The system also helps prevent mistakes during data entry. All forms include automatic checks and calendar tools to ensure the Secretary and Treasurer enter accurate details every time.

3.5 System Development Environment

The development of the KEBANA Management System requires a specific set of hardware and software tools to ensure that the project is built efficiently and meets the technical requirements outlined earlier. The selection of these tools is driven by the need for open-source availability, community support, and compatibility with the proposed "Low-Cost" hosting strategy.

3.5.1 Hardware Requirements

The development process is primarily conducted on a standard laptop workstation equipped with an Intel Core i5 processor and 8GB of RAM, which is sufficient to run the local server and database environments. For the deployment phase, the system does not require a physical server on-site. Instead, it utilizes cloud-based infrastructure. The project targets a cloud Platform-as-a-Service (PaaS) environment, specifically Render or PythonAnywhere, which provides virtualized server resources capable of handling the lightweight traffic expected from the committee's usage. This approach eliminates the need for KEBANA to purchase expensive server hardware, aligning with the project's sustainability objectives.

3.5.2 Software Requirements

The software stack is built around the Django web framework, which is written in the Python programming language. Django was selected because it follows a "batteries-included" philosophy, providing built-in security features such as protection against SQL injection and Cross-Site Scripting (XSS) out of the box. This ensures that the system remains secure without requiring extensive custom security coding. For data storage, MySQL is utilized as the Relational Database Management System (RDBMS). MySQL is chosen for its reliability in handling structured data,

such as membership records and financial transactions, and its seamless integration with Django's Object-Relational Mapper (ORM).

On the client side, the user interface is constructed using HTML5, CSS3, and JavaScript, underpinned by the Bootstrap 5 framework. Bootstrap is essential for achieving the "Mobile-First" design requirement, as its grid system automatically adjusts the layout to fit smartphone screens. For the development environment, Visual Studio Code (VS Code) serves as the Integrated Development Environment (IDE), providing tools for code debugging and version control. Finally, Draw.io is used to generate all system diagrams, ensuring that the architectural documentation remains clear and professional.

3.6 System Testing Plan

To ensure the system is reliable and functions as intended, a comprehensive testing strategy will be implemented throughout the development lifecycle. This strategy is divided into two main phases which are Unit Testing, and User Acceptance Testing (UAT).

3.6.1 Unit Testing

Unit testing involves verifying individual components of the source code to ensure they function correctly in isolation. During the development of the Financial Module, for example, specific tests will be written to check the calculation logic. If a Treasurer records an income of RM 500 and an expense of RM 100, the unit test will automatically verify that the system calculates the balance as exactly RM 400. This automated testing process helps to identify and fix logical errors early in the coding phase, preventing bugs from compounding into larger issues later in the project.

3.6.2 User Acceptance Testing (UAT)

The final and most critical phase of testing is User Acceptance Testing (UAT), which involves the actual KEBANA committee members. Once a sprint is completed, such as the Membership Database module, a UAT session will be scheduled with the Secretary and Treasurer. During this session, the committee members will be given specific tasks to perform, such as "Register a new member from your village" or "Upload a meeting minute PDF."

The goal of UAT is not just to find technical bugs, but to validate the system's usability. The developer will observe how easily the committee members interact with the interface. If a user struggles to find the "Save" button or finds the font size too small on their phone, this feedback will be documented. These insights will then be used to refine the user interface before the final system is deployed. This collaborative approach ensures that the final delivered product is not only technically sound but also practically useful for the daily operations of the KEBANA association.

3.7 Chapter Summary

In summary, this chapter has outlined the Agile methodology adopted for the KEBANA Management System, justifying its use through the need for iterative feedback from the committee. The functional requirements have been clearly defined to address the specific roles of the Secretary and Treasurer, while the system design utilizes industry-standard diagrams to map out the architecture. By utilizing a robust open-source technology stack and a rigorous testing plan, the project is well-positioned to deliver a secure, sustainable, and user-friendly digital solution that meets the objectives set out in Chapter 1.

CHAPTER 4: EXPECTED RESULTS & DISCUSSION

4.1 Introduction

This chapter outlines the anticipated results of the KEBANA Management System project. It discusses the potential impact of the proposed system on the daily operations of the association and analyzes how the digital solution addresses the problems identified in the earlier stages of this study. Furthermore, this chapter highlights the significance of the project in the context of the Sarawak Post COVID-19 Development Strategy 2030 and addresses potential challenges that may arise during the implementation phase.

4.2 Expected Outcomes

The primary expected outcome of this project is the successful deployment of a centralized membership database. This database serves as a single source of truth for the organization and eliminates the data redundancy issues caused by the previous use of multiple Excel files. We anticipate that the system will significantly reduce the time required for the Secretary to retrieve member details. Tasks that previously took several minutes of searching through physical files or scattered spreadsheets will likely be completed in a few seconds using the digital search function. This improvement in efficiency allows the committee to respond to member inquiries much faster.

In terms of event management, the system provides a structured digital archive for all association activities. We expect this feature to solve the long-standing issue of lost documents. By allowing the Secretary to upload proposal letters and meeting minutes directly to the cloud server, the system ensures that the institutional memory of KEBANA is preserved. This implies that even

if a committee member resigns or a laptop is damaged, the historical data of the association remains safe and accessible to the new committee members.

The financial module is expected to bring a new level of transparency and accountability to the association. Currently, the Treasurer relies on manual calculations which are prone to human error. The new system automates these calculations to ensure that the final balance is always accurate. We anticipate that the automated generation of monthly financial statements will simplify the auditing process during the Annual General Meeting. This feature builds greater trust between the committee and the general community because the financial records are maintained in a professional and auditable format.

4.3 Discussion of Project Significance

The significance of this project extends beyond mere administrative efficiency. It represents a crucial step towards digital inclusivity for rural organizations. By adopting this system, KEBANA aligns itself with the Sarawak Post COVID-19 Development Strategy 2030 which advocates for data-driven decision making and social inclusivity. The project demonstrates that digital transformation is not exclusive to large urban corporations but is also accessible to community-level organizations. This system serves as a model for other rural NGOs in Sarawak to modernize their operations without incurring high costs.

From a sustainability perspective, the project is designed to be financially viable for the long term. The decision to use lightweight open-source technologies ensures that the maintenance costs remain negligible. This is a critical discussion point because many digitalization projects fail due to high operating expenses. By prioritizing a low-cost architecture, the KEBANA Management

System ensures that digital transformation is sustainable and does not become a financial burden on community funds.

4.4 Potential Implementation Challenges

Despite the clear benefits, the implementation of the system may face challenges related to digital literacy. Some senior committee members may feel uncomfortable moving away from their familiar paper records and using a web-based interface. To mitigate this risk, the user interface uses a minimalist design with large text and clear labels. Additionally, the project plan includes a comprehensive training session and the provision of a printed user manual to guide the committee members through the transition.

Another potential challenge is the reliance on internet connectivity which can be unstable in certain rural areas. To address this, the system is designed to be lightweight so that it functions well even on slower 4G mobile connections. We also implemented a data export feature that allows the committee to download their data into Excel format. This ensures that they always have a local backup copy of their records in case of a prolonged internet outage or server unavailability.

4.5 Conclusion

In conclusion, the KEBANA Management System is poised to solve the critical issues of data security, administrative inefficiency, and financial opacity currently faced by the association. The technical design proves that it is possible to build a secure and modern application using cost-effective tools.

By bridging the gap between manual workflows and digital efficiency, this project fulfills its objectives and supports the broader state agenda of creating an inclusive digital society. The planning and design phases are now complete, and the project is ready to proceed to the development phase in the next semester.

REFERENCES

1. Article on NGOs and Digital Strategy

Bălăcescu, A. (2021). Visibility and communication of small NGOs in the context of digital transformation. *Journal of Research and Innovation for Sustainable Society*, 3(2), 201–208.
<https://doi.org/10.33727/JRISS.2021.2.22:201-208>

2. Article on Community Development in Malaysia/Indonesia

Haron, H., Sajari, A., Dewanti, R., Ganesan, Y., & Gui, A. (2023). Digital transformation in community development of Malaysia and Indonesia. *International Conference on Community Development*, 5(1), 232–242. <https://doi.org/10.33068/iccd.v5i1.591>

3. The Sarawak Government Strategy Document

Economic Planning Unit Sarawak. (2021). Post COVID-19 development strategy 2030. Sarawak Government.

4. WildApricot (Commercial Software)

Personify. (2024). WildApricot features: All-in-one membership management software.
<https://www.wildapricot.com/features>

5. CiviCRM (Open-Source Software)

CiviCRM. (2024). About CiviCRM: The open-source CRM for non-profits.
<https://civicrm.org/about>

6. Microsoft Excel (Issues with Data Integrity)

MemberJungle. (2023, September 18). The problem with using Excel spreadsheets to manage your club. <https://www.memberjungle.com/blog/the-problem-with-using-excel-spreadsheets-to-manage-your-club>

APPENDICES

1. Letter of Intent



**PERSATUAN KENYAH BADENG SARAWAK
(SARAWAK KENYAH BADENG ASSOCIATION) KEBANA
NO. 57, JADE GARDEN, JALAN TUN HUSSEIN ONN,
97000 BINTULU.
No. Pendaftaran: PPM-005-13-01122004**

23rd Dec 2025

Dr. Chew Kim Mey
Faculty of Computer Science and Information Technology
Universiti Malaysia Sarawak
Kota Samarahan 94300
Sarawak, Malaysia

RE: Letter of Intent for collaborating with UNIMAS FCSIT on *Digital Management System for Persatuan Kenyah Badeng Sarawak - FYP Project*

Dear *Dr. Chew Kim Mey*,

Persatuan Kenyah Badeng Sarawak (KEBANA) is pleased to confirm its intention to collaborate with the Faculty of Computer Science and Information Technology, Universiti Malaysia Sarawak (UNIMAS FCSIT) for the stated **FYP Project**, which will be carried out by **Cedric Hilary Samah** as a final year project during Semester 1 and Semester 2, 2025-2026 under your supervision.

For your information, Dr., the KEBANA committee have been working manually ever since the establishment in 2001. We at KEBANA are grateful and thankful for having one of our members of the Kenyah Badeng community, Cedric Hilary Samah, for taking these initiatives in contributing and helping us to develop the prototype for Digital Management System for KEBANA. We truly hope that this project will bring ease and efficiency in the workflow for the KEBANA upcoming events. During this period, he will be co-mentored by Mr. Reniu Sageng, the Secretary of the Persatuan KEBANA.

Both parties agree not to disclose any confidential information concerning the business affairs or products of the other, which may be acquired during the activities pertaining to this project.

We look forward to working with UNIMAS FCSIT.

Sincerely,

Reniu Sageng
Secretary
Persatuan Kenyah Badeng Sarawak (KEBANA)